Table B4. Spline-indicator fixed-parameter binary logit model of crash injury severity (serious injury/fatal vs. PDO/minor injury), multi-vehicle crashes without a motorcycle, Thailand, 2016–2023: implied parameter for each year, estimated parameter (*t*-statistic).

Variable description	Spline terms retained (*j*)	2016	2017	2018	2019	2020	2021	2022	2023
Constant [serious injury / fatal]									
Constant	1, 2, 8	–0.501 (–4.07)	–0.600 (–5.83)	–0.699 (–7.68)	–0.798 (–8.79)	–0.897 (–8.78)	–0.996 (–8.16)	–1.095 (–7.44)	–0.668 (–3.71)
Vehicle involvement									
One passenger car involved (1 if yes, 0 otherwise)	1, 3, 5	–0.388 (–8.44)	–0.777 (–8.44)	–0.577 (–9.69)	–0.377 (–5.12)	–0.424 (–7.46)	–0.471 (–8.74)	–0.518 (–7.77)	–0.565 (–6.38)
Two or more passenger cars involved (1 if yes, 0 otherwise)	1, 2, 3, 5	–1.291 (–9.00)	–1.534 (–10.31)	–1.197 (–12.50)	–0.861 (–7.54)	–0.980 (–10.85)	–1.100 (–12.20)	–1.220 (–10.73)	–1.340 (–8.93)
One pickup truck involved (1 if yes, 0 otherwise)	1, 3, 7	–0.209 (–5.12)	–0.418 (–5.12)	–0.430 (–7.03)	–0.443 (–8.61)	–0.456 (–7.87)	–0.469 (–6.12)	–0.340 (–4.98)	–0.212 (–2.13)
Two or more pickup trucks involved (1 if yes, 0 otherwise)	1, 3	–0.541 (–10.05)	–1.081 (–10.05)	–1.035 (–11.88)	–0.989 (–13.39)	–0.943 (–13.09)	–0.896 (–10.89)	–0.850 (–8.40)	–0.804 (–6.44)
One 6-wheel truck involved (1 if yes, 0 otherwise)	1, 7, 8	–0.016 (–1.01)	–0.032 (–1.01)	–0.048 (–1.01)	–0.065 (–1.01)	–0.081 (–1.01)	–0.097 (–1.01)	0.217 (1.60)	0.005 (0.04)
One 10-wheel truck involved (1 if yes, 0 otherwise)	1	0.009 (0.78)	0.018 (0.78)	0.027 (0.78)	0.036 (0.78)	0.045 (0.78)	0.054 (0.78)	0.063 (0.78)	0.072 (0.78)
One truck with more than 10 wheels involved (1 if yes, 0 otherwise)	1, 4	0.088 (3.31)	0.176 (3.31)	0.264 (3.31)	0.241 (3.89)	0.218 (4.12)	0.195 (3.43)	0.171 (2.40)	0.148 (1.61)
Two or more trucks with more than 10 wheels involved (1 if yes, 0 otherwise)	1, 3	–0.219 (–2.45)	–0.437 (–2.45)	–0.397 (–2.79)	–0.357 (–3.07)	–0.317 (–2.94)	–0.277 (–2.29)	–0.237 (–1.58)	–0.197 (–1.05)
Van involved (1 if yes, 0 otherwise)	1, 2, 3	0.295 (1.74)	–0.225 (–1.80)	–0.158 (–1.60)	–0.090 (–1.13)	–0.023 (–0.32)	0.044 (0.52)	0.111 (1.05)	0.178 (1.33)
Bus involved (1 if yes, 0 otherwise)	1, 6, 7	–0.027 (–0.78)	–0.054 (–0.78)	–0.081 (–0.78)	–0.108 (–0.78)	–0.135 (–0.78)	0.604 (1.75)	–0.032 (–0.16)	–0.667 (–2.20)
Crash type									
Rear-end collision in the same lane (1 if yes, 0 otherwise)	1, 2, 6, 7	–0.303 (–3.62)	–0.330 (–5.41)	–0.357 (–7.06)	–0.384 (–6.48)	–0.411 (–5.06)	–0.169 (–1.88)	–0.279 (–3.78)	–0.388 (–3.56)
Sideswipe (parallel movement) (1 if yes, 0 otherwise)	1, 2, 5, 6	–0.574 (–3.79)	–0.630 (–6.27)	–0.686 (–6.95)	–0.742 (–5.01)	–0.259 (–2.09)	–0.349 (–4.08)	–0.439 (–4.89)	–0.528 (–3.98)
Head-on collision (1 if yes, 0 otherwise)	1, 3	0.528 (9.26)	1.056 (9.26)	1.006 (11.22)	0.957 (13.49)	0.907 (14.39)	0.857 (12.30)	0.807 (9.20)	0.757 (6.78)
Crash cause									
Speeding (1 if yes, 0 otherwise)	1, 2, 3, 5	–0.175 (–1.74)	0.255 (2.78)	0.053 (0.76)	–0.149 (–1.72)	–0.073 (–1.06)	0.002 (0.04)	0.078 (0.95)	0.154 (1.44)
Cutting in too closely (1 if yes, 0 otherwise)	1, 7, 8	–0.043 (–2.32)	–0.086 (–2.32)	–0.129 (–2.32)	–0.172 (–2.32)	–0.215 (–2.32)	–0.258 (–2.32)	0.166 (1.12)	–0.286 (–1.76)
Driver fell asleep (1 if yes, 0 otherwise)	1, 3, 5	0.439 (4.73)	0.878 (4.73)	0.548 (4.57)	0.219 (1.37)	0.301 (2.49)	0.383 (3.54)	0.464 (3.58)	0.546 (3.15)
Roadway characteristics									
Crash on the main carriageway (1 if yes, 0 otherwise)	1, 2, 7	–0.598 (–4.73)	–0.409 (–4.29)	–0.220 (–2.95)	–0.031 (–0.42)	0.159 (1.71)	0.348 (2.81)	0.112 (1.19)	–0.123 (–0.79)
Road under construction (1 if yes, 0 otherwise)	1, 5, 6, 8	0.217 (4.00)	0.434 (4.00)	0.652 (4.00)	0.869 (4.00)	–0.236 (–0.81)	0.114 (0.71)	0.464 (2.34)	–0.121 (–0.54)
Painted median (1 if yes, 0 otherwise)	1, 5, 7, 8	0.104 (3.63)	0.209 (3.63)	0.313 (3.63)	0.417 (3.63)	0.138 (1.50)	–0.140 (–0.90)	0.258 (1.43)	–0.136 (–0.67)
Raised or earth median (1 if yes, 0 otherwise)	1, 2, 6, 7, 8	–0.602 (–5.88)	–0.464 (–6.14)	–0.327 (–5.29)	–0.189 (–2.74)	–0.051 (–0.55)	–0.390 (–3.57)	–0.167 (–1.48)	–0.601 (–4.60)
Depressed median (1 if yes, 0 otherwise)	1, 2, 4, 6	–0.408 (–3.83)	–0.426 (–5.89)	–0.443 (–4.65)	–0.250 (–3.58)	–0.057 (–0.61)	–0.172 (–2.48)	–0.288 (–3.57)	–0.404 (–3.48)
Barrier median (1 if yes, 0 otherwise)	1, 2, 4, 5	–1.447 (–9.12)	–1.054 (–10.40)	–0.661 (–4.90)	–1.375 (–11.16)	–1.483 (–16.28)	–1.592 (–20.00)	–1.701 (–17.66)	–1.809 (–13.83)
Horizontal curve (1 if yes, 0 otherwise)	1, 2	0.289 (2.85)	0.209 (2.55)	0.129 (1.95)	0.049 (0.86)	–0.031 (–0.52)	–0.110 (–1.60)	–0.190 (–2.22)	–0.270 (–2.56)
Concrete pavement surface (1 if yes, 0 otherwise)	1	–0.017 (–1.83)	–0.035 (–1.83)	–0.052 (–1.83)	–0.069 (–1.83)	–0.087 (–1.83)	–0.104 (–1.83)	–0.121 (–1.83)	–0.139 (–1.83)
On grade (1 if yes, 0 otherwise)	1, 2	–0.460 (–2.39)	–0.402 (–2.55)	–0.343 (–2.68)	–0.285 (–2.61)	–0.227 (–2.13)	–0.168 (–1.39)	–0.110 (–0.75)	–0.052 (–0.28)
T-intersection (1 if yes, 0 otherwise)	1, 3, 4	0.101 (1.10)	0.202 (1.10)	–0.300 (–1.86)	–0.276 (–2.26)	–0.251 (–2.69)	–0.226 (–2.63)	–0.202 (–1.93)	–0.177 (–1.28)
Four-leg intersection (1 if yes, 0 otherwise)	1, 2, 4, 5	–0.354 (–2.08)	0.046 (0.42)	0.446 (2.70)	–0.287 (–1.95)	–0.257 (–2.43)	–0.226 (–2.54)	–0.196 (–1.79)	–0.165 (–1.08)
U-turn with deceleration lane only (1 if yes, 0 otherwise)	1, 4, 6	0.220 (2.74)	0.440 (2.74)	0.660 (2.74)	0.240 (1.60)	–0.181 (–0.85)	–0.000 (–0.00)	0.181 (1.23)	0.361 (1.59)
U-turn with deceleration and acceleration lanes (1 if yes, 0 otherwise)	1, 2, 5	0.467 (1.86)	0.266 (1.59)	0.065 (0.43)	–0.136 (–0.62)	0.021 (0.13)	0.178 (1.23)	0.334 (1.73)	0.491 (1.80)
Environmental characteristics									
Wet road surface (1 if yes, 0 otherwise)	1, 2, 3, 4, 6	–0.336 (–0.84)	–22.557 (–46.31)	–1.370 (–2.03)	–0.685 (–1.92)	–0.001 (–0.00)	–0.208 (–0.99)	–0.415 (–1.82)	–0.622 (–1.78)
Raining (1 if yes, 0 otherwise)	1, 2, 3, 4, 5, 6	0.149 (0.35)	22.417 (46.02)	0.838 (1.22)	0.652 (1.70)	–0.426 (–1.29)	–0.241 (–1.09)	–0.057 (–0.23)	0.128 (0.35)
Dark, no street lighting (1 if yes, 0 otherwise)	1, 2, 5, 6	0.777 (4.67)	0.931 (7.01)	1.084 (8.21)	1.237 (7.52)	0.412 (2.61)	0.441 (3.29)	0.469 (3.11)	0.498 (2.51)
Dark, with street lighting (1 if yes, 0 otherwise)	1, 3	0.291 (3.78)	0.582 (3.78)	0.491 (3.97)	0.400 (3.90)	0.309 (3.18)	0.218 (1.98)	0.127 (0.94)	0.036 (0.21)
Temporal characteristics									
Weekend (1 if yes, 0 otherwise)	1	0.015 (2.11)	0.029 (2.11)	0.044 (2.11)	0.059 (2.11)	0.074 (2.11)	0.088 (2.11)	0.103 (2.11)	0.118 (2.11)
Songkran festival period (1 if yes, 0 otherwise)	1	0.000 (0.03)	0.001 (0.03)	0.001 (0.03)	0.002 (0.03)	0.002 (0.03)	0.002 (0.03)	0.003 (0.03)	0.003 (0.03)
New Year festival period (1 if yes, 0 otherwise)	1, 2, 3, 5, 6	–0.497 (–3.06)	0.245 (1.56)	–0.094 (–0.93)	–0.433 (–2.72)	–0.006 (–0.04)	–0.126 (–1.22)	–0.246 (–2.27)	–0.367 (–2.19)
Evening (1 if yes, 0 otherwise)	1, 2, 3, 8	0.262 (1.85)	–0.263 (–1.60)	–0.126 (–0.96)	0.010 (0.09)	0.147 (1.38)	0.283 (2.29)	0.420 (2.73)	0.270 (1.41)

Variable description	Spline terms retained (*j*)	2016	2017	2018	2019	2020	2021	2022	2023
Midnight / early morning (1 if yes, 0 otherwise)	1, 2, 3	0.487 (3.38)	0.198 (1.32)	0.254 (2.12)	0.310 (3.12)	0.366 (3.87)	0.422 (3.92)	0.478 (3.58)	0.534 (3.22)
Peak hours (1 if yes, 0 otherwise)	1	0.018 (1.99)	0.036 (1.99)	0.054 (1.99)	0.072 (1.99)	0.091 (1.99)	0.109 (1.99)	0.127 (1.99)	0.145 (1.99)
Region									
Eastern region (1 if yes, 0 otherwise)	1, 2	0.376 (3.81)	0.397 (4.91)	0.417 (6.38)	0.438 (7.94)	0.459 (8.66)	0.480 (8.02)	0.500 (6.85)	0.521 (5.79)
Northeastern region (1 if yes, 0 otherwise)	1, 2, 3, 7	0.509 (4.45)	0.311 (3.42)	0.427 (6.42)	0.544 (9.97)	0.660 (10.60)	0.777 (9.19)	0.566 (8.04)	0.355 (3.09)
Northern region (1 if yes, 0 otherwise)	1, 5	0.016 (0.58)	0.032 (0.58)	0.049 (0.58)	0.065 (0.58)	–0.057 (–0.69)	–0.179 (–2.24)	–0.301 (–2.84)	–0.424 (–2.90)
Southern region (1 if yes, 0 otherwise)	1, 6	0.113 (6.25)	0.226 (6.25)	0.339 (6.25)	0.452 (6.25)	0.566 (6.25)	0.387 (5.72)	0.208 (2.50)	0.029 (0.24)
Western region (1 if yes, 0 otherwise)	1, 2, 3, 6	0.434 (2.75)	0.036 (0.25)	0.307 (3.17)	0.577 (6.87)	0.848 (7.15)	0.663 (7.80)	0.478 (4.81)	0.293 (1.98)
Model statistics									
Number of observations					33,646				
Number of estimated spline coefficients					137				
Variables with the base term only (parameter proportional to the period index)					5 of 45				
Log-likelihood at zero, LL(0)					–23321.63				
Log-likelihood with constant only, LL(C)					–15177.81				
Log-likelihood at convergence, LL(β)					–12653.02				
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$					0.457				
AIC / BIC					25,580.0 / 26,734.1				
LR test against the fixed-parameter separate-period models					$\chi^2 = 139.69$, df = 223, confidence of rejection = 0.000				

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median. Following Shabab et al. (2024), each variable enters through spline terms $f_j(\tau) = \max(\tau - j + 1, 0)$, $j = 1, \ldots, 8$, where $\tau = 1, \ldots, 8$ indexes the years; the parameter in year τ is $\beta(\tau) = \sum_j b_j f_j(\tau)$. The base term ($j = 1$) is always retained; the other terms were removed by backward elimination at the 90% level ($|t| < 1.645$, one term per variable per round). A variable with the base term only has a parameter proportional to the period index. The table reports the implied parameters $\beta(\tau)$; *t*-statistics by the delta method from the covariance of the spline coefficients.